

Informational Requirements for Pre-Categorical Relational Models: Identifiability, Measure, Dynamic Closure, and Coarse-Graining

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Technical Abstract

This technical research report studies relational models that are asked to support cardinal comparison, refinement, transition, and coarse-graining even when their primitive data contain only a relation. We study which additional informational structures are required for those operations. Order-only observations are invariant under broad monotone recalibrations and therefore do not identify latent marginal scales or densities. An additional mark q can supply relative cardinal information only when its marked law and observable coupling are not determined by the relation. Additive mass μ is required to distinguish conservative refinement from representational cloning, while locality and symmetry alone leave the transition kernel K underdetermined. For the exponential observable, the block message consisting of total mass and

$$Q_\lambda = -\lambda^{-1} \log \left(\frac{\sum_i \mu_i e^{-\lambda q_i}}{\sum_i \mu_i} \right)$$

aggregates exactly. Under the centered contraction $q'_i = \bar{q}_\mu + a(q_i - \bar{q}_\mu)$, however,

$$Q'_\lambda = (1 - a)\bar{q}_\mu + aQ_{a\lambda}.$$

We give an exact four-point counterexample showing that equal mean and equal Q_λ do not determine the next score on the unrestricted finite class. Thus exact static sufficiency does not imply dynamic autonomy. For Markov coarse-graining, a microscopic kernel and a partition do not generally determine a macro kernel without source occupancy π or lumpability; joint flow aggregates exactly when occupancy is supplied. These obstructions motivate the provisional informational architecture (R, q, μ, K, π) . The construction is formal and does not derive physical spacetime, calibration, or a fundamental dynamics.

Keywords

Relational models; identifiability; measurement theory; coarse-graining; dynamic closure; Markov aggregation

1. Introduction

Relational models often begin with a sparse primitive: a set of entities and a relation among them. From that starting point, one may wish to define cardinal comparisons, assign multiplicities, specify transition probabilities, or construct macroscopic descriptions. These operations are not interchangeable. Each requires information that may or may not be present in the original relation.

This report studies that dependency problem. The question is not whether one particular relational model is physically correct. It is:

Which informational structures must be supplied before a relational description can support the quantitative and dynamical operations assigned to it?

In this report, **pre-categorical** denotes the informational stage before a relational description is assigned a fixed metric, measure-theoretic, dynamical, or physical interpretation. The expression is not used in the technical sense of category theory.

This report belongs to the **Modal Field Research Programme**, a broader obstruction-driven investigation of relational modelling. No specific physical interpretation from that wider programme is assumed here.

The method is obstruction-driven. A new component is introduced only after an invariance argument, identifiability result, underdetermination construction, or closure counterexample shows that the previous representation is insufficient for a declared operation. The resulting inventory is

$$\mathcal{M} = (R, q, \mu, K, \pi),$$

where R is relational or ordinal structure, q is an additional cardinal mark, μ is additive mass, K is a conditional transition kernel, and π is source occupancy for dynamic aggregation. This tuple is a provisional informational architecture. It is not asserted to be a unique ontology or a derivation of physical spacetime.

1.1 Terminology and notation

The principal symbols used throughout the report are collected here for reference.

Symbol	Role in the report
R	observed relational or ordinal structure
q	additional cardinal mark
μ	positive additive mass used for refinement and aggregation
K	conditional transition kernel
π	source occupancy used in dynamic aggregation
λ	parameter of the exponential aggregate Q_λ

Symbol	Role in the report
β	strength parameter of a declared local exponential kernel
Q_λ	mass-normalized effective exponential score
z	standardized contextual score, defined only on a nondegenerate domain

The first obstruction concerns order-only identifiability. Strict coordinatewise monotone transformations can preserve an induced finite order while changing latent marginal scales and densities. Under continuous marginals, the law of a coordinatewise order depends at most on the copula, not on the marginal calibrations. Consequently, a deterministic order-only estimator cannot recover targets that vary inside the same order-equivalence class.

A separate mark q can add cardinal information only when its marked law is not itself determined by R . Operational meaning additionally requires a declared observation kernel sensitive to gauge-invariant q -contrasts. Locality, relabeling equivariance, and positive-affine standardization do not select a unique kernel: an infinite family remains. Classical difference-based odds and maximum-entropy arguments can select an exponential family under stronger assumptions, but they do not determine a unique physical coupling strength.

Multiplicity creates a second obstruction. Relation plus marks cannot distinguish a descriptive refinement from the addition of ontically distinct copies when both generate the same marked representation. An additive mass μ supplies the missing refinement semantics. Once masses are supplied, conservative cloning, finite regrouping, and projective branch fractions can be handled exactly; consistency constrains how weights combine but not which weights should be used.

For a fixed parameter λ , the additive exponential weight

$$W_\lambda(B) = \sum_{i \in B} \mu_i e^{-\lambda q_i}$$

admits the exact block message

$$(M_B, Q_\lambda(B)), \quad M_B = \sum_{i \in B} \mu_i,$$

where

$$Q_\lambda(B) = -\lambda^{-1} \log \left(\frac{W_\lambda(B)}{M_B} \right).$$

This is classical log-sum-exp algebra applied to the present architecture. Its exact static associativity does not imply autonomous macrodynamics. Under the centered contraction

$$q'_i = \bar{q}_\mu + a(q_i - \bar{q}_\mu),$$

affine covariance gives

$$Q'_\lambda = (1 - a)\bar{q}_\mu + aQ_{a\lambda}.$$

The report provides an exact four-point counterexample showing that two finite distributions can have the same mass-normalized mean and the same Q_λ , yet different $Q_{a\lambda}$ and therefore different next Q_λ . Thus one fixed- λ score is not a generally closed dynamic state.

A third distinction appears in coarse-graining. The exact message depends on the observable to be preserved. Arithmetic means preserve first moments; exponential scores preserve exponential weights; other targets require other messages. For Markov dynamics, a microscopic kernel and a partition do not generally determine a unique macro kernel without source occupancy or a lumpability condition. Carrying the joint flow $F_{ij} = \pi_i K_{ij}$ restores exact aggregation.

The contribution is therefore a structured separation of informational roles. Several ingredients are classical: copula invariance, measurement-scale arguments, exponential choice models, additive measure extension, quasi-arithmetic means, and Markov lumpability. The project-specific value lies in the obstruction map that organizes them, the explicit separation of static sufficiency from dynamic autonomy, and the exact nonclosure witness for the chosen centered contraction.

The report proceeds as follows. Section 2 establishes the order-only identifiability limit. Sections 3 and 4 introduce q , its gauge contract, and the status of operational kernels and their parameters. Section 5 treats measure, multiplicity, and refinement. Sections 6 and 7 separate exact static aggregation from dynamic closure. Section 8 develops observable-relative coarse-graining and occupancy-weighted flow aggregation. Section 9 assembles the provisional architecture. Section 10 states the scope, contribution, and open bridges. Appendices record equivariance, boundary cases, restricted closure, transform-domain conditions, and a nondecomposable median example.

2. Order-Only Identifiability

2.1 Relational observation

Let $I = \{1, \dots, n\}$ be a finite labeled set. A relational observation is a binary relation

$$R \subseteq I \times I.$$

The discussion in this section applies most directly when R is obtained from a latent coordinatewise order. Let

$$Z_i = (X_i^{(1)}, \dots, X_i^{(d)}) \in \mathbb{R}^d$$

and define

$$i \preceq_R j \iff X_i^{(r)} \leq X_j^{(r)} \text{ for every } r = 1, \dots, d.$$

The relation records which labeled pairs are comparable in every coordinate. It does not record the numerical coordinate values, the units used for those coordinates, or the marginal sampling densities. This distinction is not a defect of the representation: it is the information content of the representation.

The central question is therefore an identifiability question. Which latent quantities are constant on every class of coordinate models that produce the same observed relation? Any quantity that varies inside such a class cannot be recovered by a deterministic estimator

whose input is only R and n .

2.2 Monotone recalibration

Lemma 2.1 (coordinatewise monotone invariance).

For each coordinate r , let

$$T_r : \mathbb{R} \rightarrow \mathbb{R}$$

be strictly increasing. Set

$$\tilde{X}_i^{(r)} = T_r(X_i^{(r)}).$$

Then the coordinatewise order generated by $\tilde{Z}_i = (\tilde{X}_i^{(1)}, \dots, \tilde{X}_i^{(d)})$ is identical to the order generated by Z_i .

Proof. Strict monotonicity gives

$$X_i^{(r)} \leq X_j^{(r)} \iff T_r(X_i^{(r)}) \leq T_r(X_j^{(r)})$$

for every coordinate r . Taking the conjunction over all coordinates preserves every relation entry. Hence the two labeled relation matrices are equal. $\square$

The lemma is elementary, but its consequence is substantial. The transformations T_r can strongly alter coordinate spacings and marginal densities while leaving the observed order unchanged. Therefore numerical density, marginal scale, and coordinate spacing are not functions of the order alone.

This statement should not be confused with the stronger claim that order contains no geometric information. An order can encode rich invariant structure. The point is narrower: information that changes under order-preserving recalibration is not identifiable from the order without an additional convention or observable.

2.3 Copula dependence

Suppose now that the latent vectors are independent and identically distributed, and consider the two-dimensional case for clarity. Let (X_i, Y_i) , $i = 1, \dots, n$, have continuous strictly increasing marginal distribution functions F_X and F_Y . Define

$$i \prec_R j \iff X_i < X_j \text{ and } Y_i < Y_j.$$

Set

$$U_i = F_X(X_i), \quad V_i = F_Y(Y_i).$$

By strict monotonicity,

$$X_i < X_j \iff U_i < U_j, \quad Y_i < Y_j \iff V_i < V_j$$

for every realized sample. The probability-integral transform therefore couples the original sample to a sample with uniform marginals and exactly the same labeled order. By Sklar's theorem, the joint law of (U, V) is the copula of (X, Y) (Sklar 1959).

Proposition 2.2 (copula sufficiency for the induced order law).

Under continuous strictly increasing marginals, the probability law of the finite labeled coordinatewise order depends on the latent joint distribution only through its copula.

Proof. Under the coupling above, the complete labeled relation matrix constructed from (X_i, Y_i) is pointwise identical to the matrix constructed from (U_i, V_i) . The latter has uniform marginals and the same copula. Consequently every event measurable from the finite order has a probability determined by the copula. $\square$

The proposition says “depends at most on the copula.” It does not assert that different copulas always induce different order laws at every finite sample size. That converse would require a separate injectivity result.

2.4 An order-only identifiability limit

Let $\mathcal{M}$ be a class of latent models, and let

$$\Phi : \mathcal{M} \rightarrow \mathcal{R}_n$$

map a latent model to its observed labeled relation. Let $\theta : \mathcal{M} \rightarrow \Theta$ be a target quantity.

Proposition 2.3 (fiberwise identifiability criterion).

A deterministic estimator $T : \mathcal{R}_n \rightarrow \Theta$ satisfying

$$T(\Phi(M)) = \theta(M) \quad \text{for every } M \in \mathcal{M}$$

can exist only if θ is constant on every fiber of Φ .

Proof. If $\Phi(M_1) = \Phi(M_2)$, then determinism implies

$$T(\Phi(M_1)) = T(\Phi(M_2)).$$

Therefore exact recovery requires $\theta(M_1) = \theta(M_2)$. $\square$

Combining this criterion with Lemma 2.1 yields the order-only obstruction: marginal scales, marginal densities, and any other quantities altered by strict coordinatewise recalibration cannot be recovered from a single finite relation matrix and its cardinality. Proposition 2.2 extends the conclusion to repeated independent samples: two models with the same copula and different continuous marginals induce the same law for every finite sequence of observed orders, so order-only statistical procedures cannot consistently distinguish those marginal models.

This is an identifiability statement, not a universal impossibility claim. The distinction between qualitative structure, numerical representation, and admissible recalibration is standard in representational measurement theory (Krantz et al. 1971). The obstruction can be removed by enriching the observation. Examples include a calibrated coordinate, an independently specified measure, physically distinguished marks, repeated observations containing non-order data, or a dynamical rule whose observables are not invariant under the same recalibrations.

2.5 Consequence for relational model design

The order R should be treated as one informational layer rather than a complete state description. It can support ordinal questions—comparability, reachability, precedence,

intervals, and quotient structure—but it does not supply a preferred cardinal scale or sampling measure.

This motivates the separation adopted in the remainder of the report. A quantitative mark q is introduced only when cardinal contrasts are required; an additive mass μ is introduced only when refinement and multiplicity must be tracked; a transition kernel K is introduced only when stochastic evolution is specified; and an occupancy π is introduced only when microscopic transitions must be aggregated dynamically. None of these structures follows merely from renaming the order.

The conclusion is methodological:

Methodological criterion. Before deriving an observable from R , test whether it is constant on the order-equivalence classes.

If it is not, the observable requires additional structure.

3. Additional Quantitative Structure q

3.1 Why an additional mark is introduced

Section 2 established an identifiability limit: quantities that vary under strict order-preserving recalibrations are not recoverable from the relation R alone. A natural response is to enrich the relational state with a quantitative mark. Let $\mathcal{A}$ denote the relevant relational carriers, such as states, edges, or admissible transitions, and introduce

$$q : \mathcal{A} \rightarrow \mathbb{R}.$$

The mark q is not defined here as a function of the order. It is supplied as additional information. This distinction is essential. If q were constructed deterministically from R , then it could not distinguish two latent representatives that produce the same relation.

The same conclusion holds if the construction uses model-independent randomness. Suppose

$$q = F(R, U),$$

where U has the same conditional law in every latent model sharing the same observed relation. Then the conditional law of q is also the same throughout that order-equivalence class. Such a mark may be useful as a randomized relational statistic, but it does not add information about which latent representative generated the relation.

Proposition 3.1 (information criterion for an additional mark).

A mark q can break an order-only identifiability class only if its conditional marked law is not determined by R and differs between at least some latent models that induce the same relation.

Proof. If the conditional law of q given R is identical in two latent models, then the joint observable law of (R, q) is identical in those models. No estimator based only on (R, q) can distinguish them. Therefore distinguishability requires a conditional marked law that varies inside the order-equivalence class. $\square$

The proposition is an information statement, not a physical-origin theorem. It says what an informative q must do; it does not explain how nature, an experiment, or a deeper theory would supply it.

3.2 Operational meaning requires a coupling

A numerical label is not operational merely because it is stored in the state. Relative to an observation class $\mathcal{O}$, an operational mark requires a specified conditional law

$$K(O \mid R, q), \quad O \in \mathcal{O},$$

that is sensitive to at least some physically or statistically admissible changes in q .

If

$$K(O \mid R, q) = K(O \mid R, q')$$

for every allowed pair q, q' , then q is empirically silent for that observation class. Conversely, sensitivity alone is not enough: the coupling should respect the symmetries assigned to the representation. In the current framework, the minimum operational contract is:

1. relabeling equivariance;
2. invariance under declared gauge transformations;
3. a stated relational locality condition;
4. nontrivial dependence on gauge-invariant q -contrasts;
5. inferability of at least some q -information from observable frequencies under a calibrated model.

This contract separates two questions that are often merged. The first is whether q contains information not already encoded by R . The second is whether a declared observable law makes any of that information measurable. A mark can satisfy the first condition and fail the second.

3.3 Global-shift redundancy

Many candidate constructions use only differences, centered values, or normalized functions of q . In such cases the transformation

$$q_\alpha \mapsto q_\alpha + c$$

for one constant c changes no observable contrast.

Define the centered mark

$$\tilde{q}_\alpha = q_\alpha - \bar{q}, \quad \bar{q} = \frac{1}{|\mathcal{A}|} \sum_{\alpha \in \mathcal{A}} q_\alpha$$

for a finite carrier set. Then

$$\widetilde{(q + c)}_\alpha = \tilde{q}_\alpha.$$

Likewise, every difference satisfies

$$(q_\alpha + c) - (q_\beta + c) = q_\alpha - q_\beta.$$

Proposition 3.2 (global-shift quotient).

Any construction that factors through the set of q -differences or through the centered mark

is a function of the equivalence class

$$[q] = \{q + c\mathbf{1} : c \in \mathbb{R}\}.$$

It cannot identify an absolute additive zero for q .

The proposition is definitional. Calling the redundancy a gauge contract is useful because it prevents later formulas from depending accidentally on an unobservable offset. It does not establish a fundamental gauge symmetry of nature.

For a local normalized transition law, the redundancy can be larger. If each source row is normalized separately, adding a source-dependent constant

$$q_{ij} \mapsto q_{ij} + c_i$$

may leave the entire row unchanged. The exact gauge group therefore belongs to the declared observable kernel, not to the symbol q in isolation.

3.4 Offset and scale are different problems

Removing an additive zero does not fix a multiplicative unit. Under

$$q \mapsto aq + c, \quad a > 0,$$

differences transform as

$$q_\alpha - q_\beta \mapsto a(q_\alpha - q_\beta).$$

Thus a contrast can be shift-invariant while remaining scale-covariant. A dimensionless observable can be formed by combining q with a coupling parameter or by standardizing the local values, but those choices have different informational meanings.

A raw exponential factor,

$$e^{-\beta q_\alpha},$$

depends only on the product βq_α . Rescaling q while inversely rescaling β leaves the factor unchanged. Before a q -unit is fixed, the numerical magnitude of β is therefore inseparable from the scale of q .

Local standardization instead defines, when the local variance is positive,

$$z_\alpha = \frac{q_\alpha - \bar{q}_{\text{local}}}{s_{\text{local}}}.$$

The standardized values are invariant under positive affine recalibrations of the local raw marks. This removes raw scale from the input, but it does not select which function of z should govern an observable. That underdetermination is the subject of Section 4.

3.5 Relative information versus calibrated quantity

The current role of q is therefore limited but precise. It can provide relative cardinal information that is absent from R , provided its marked law is not itself determined by R . Gauge-invariant contrasts can become operational through a specified observation kernel.

None of these statements supplies:

- an absolute additive zero;
- a physical unit;
- a calibration procedure;
- an empirical measurement map;
- a derivation of the mark from a deeper dynamics.

A model may consistently work on the quotient space of marks modulo global shift while leaving the scale to be estimated. Alternatively, an external standard may fix both offset and unit. These are different modelling contracts and should not be blended.

The section's conclusion is:

Conclusion. The mark q is an additional informational primitive only to the extent that its marked law and observable coupling add information beyond R .

The introduction of q resolves an order-only insufficiency. It does not resolve measure, kernel selection, dynamic closure, or physical calibration.

4. Operational Kernels and Parameter Status

4.1 A local exponential witness

Let $A_{ij} \in \{0, 1\}$ indicate which transitions from state i to state j are admissible, and let q_{ij} be a mark on those transitions. A simple local observation model is

$$K_\beta(i, j) = \frac{A_{ij}e^{-\beta q_{ij}}}{\sum_k A_{ik}e^{-\beta q_{ik}}},$$

whenever the denominator is positive. This is a multinomial-logit or softmax kernel (Luce 1959; McFadden 1974).

For two admissible destinations j and k from the same source,

$$\log \frac{K_\beta(i, j)}{K_\beta(i, k)} = -\beta(q_{ij} - q_{ik}).$$

Thus observable log-odds recover local q -contrasts when β is known and sufficient transition data are available. The identity provides a mathematical witness that q -contrasts can be operationalized without assigning meaning to the common row offset.

The kernel is not derived from the relational primitives. It is one declared coupling. Its value is to make the inferential contract explicit.

Throughout this report, β denotes the strength of a declared transition kernel, while λ denotes the parameter of the exponential aggregation transform. They may be set equal in a model that intentionally uses one exponential law for both purposes, but no such identification is assumed by notation alone.

4.2 Gauge and raw-scale degeneracy

For any source-dependent shift c_i ,

$$q_{ij} \mapsto q_{ij} + c_i$$

multiplies every unnormalized weight in row i by the same factor $e^{-\beta c_i}$, which cancels under normalization. Therefore the local kernel identifies only within-row contrasts.

There is also a raw-scale degeneracy. For $a > 0$,

$$K_\beta(i, j; q) = K_{\beta/a}(i, j; aq + c_i).$$

Without a calibrated unit for q , the data identify the products $\beta(q_{ij} - q_{ik})$, not β and the raw q -scale separately.

Lemma 4.1 (raw-score reparameterization).

In the local exponential kernel, positive rescaling of the raw score can be exactly compensated by inverse rescaling of the coupling strength.

This is a standard parameterization degeneracy. It should not be described as evidence that every version of β or λ is pure gauge. Once a standardization convention fixes the admissible score representation, the remaining parameter can become identifiable within that model.

4.3 Scale-free local kernels remain underdetermined

For each source i , suppose the outgoing marks are nonconstant and define

$$z_{ij} = \frac{q_{ij} - \bar{q}_i}{s_i}.$$

For any strictly positive function $f : \mathbb{R} \rightarrow (0, \infty)$, define

$$K_f(i, j) = \frac{A_{ij}f(z_{ij})}{\sum_k A_{ik}f(z_{ik})}.$$

This family is local, relabeling-equivariant, and invariant under every row-wise positive affine recalibration

$$q_{ij} \mapsto a_i q_{ij} + b_i, \quad a_i > 0.$$

Yet different positive functions f generally produce different transition probabilities on the same marked graph.

Proposition 4.2 (kernel underdetermination).

Relational locality, relabeling equivariance, and positive-affine invariance of standardized local scores do not select a unique transition law. They admit an infinite family K_f .

The proposition blocks a common shortcut. Removing units and enforcing symmetry can restrict the admissible representation, but those requirements alone do not derive an exponential kernel or any other unique physical law.

4.4 What IIA does and does not select

For a finite choice set S , every normalized pointwise-weight model

$$P(j \mid S) = \frac{f(z_j)}{\sum_{k \in S} f(z_k)}$$

satisfies

$$\frac{P(j | S)}{P(k | S)} = \frac{f(z_j)}{f(z_k)}.$$

The pairwise odds do not depend on alternatives outside the pair. Thus the normalized pointwise representation has the independence-of-irrelevant- alternatives property associated with Luce's choice framework (Luce 1959). IIA by itself does not force f to be exponential.

A stronger condition does. Suppose the odds ratio depends only on score difference:

$$\frac{f(x)}{f(y)} = g(x - y),$$

where f and g are positive and continuous. Setting $y = 0$ expresses f through g , and comparison of three scores gives

$$g(a + b) = g(a)g(b).$$

The positive continuous solutions are

$$g(t) = e^{-\lambda t},$$

so

$$f(z) = Ce^{-\lambda z}.$$

Proposition 4.3 (difference-based exponential selection).

A positive continuous pointwise choice representation whose odds depend only on score difference belongs to the exponential family.

This is a classical functional-equation result within the Luce/logit tradition. The project uses it as an attributed selection principle. It is not a new derivation of multinomial logit, and it does not determine the numerical value of λ .

4.5 Maximum entropy selects a family conditional on a constraint

A second route begins with an optimization problem. Let z_j be supplied scores and maximize Shannon entropy

$$H(p) = - \sum_j p_j \log p_j$$

subject to

$$\sum_j p_j = 1, \quad \sum_j p_j z_j = m.$$

The Lagrange stationary condition gives

$$p_j \propto e^{-\lambda z_j}.$$

Strict concavity yields a unique interior maximizer for a feasible constraint. With a reference mass μ_j , maximizing relative entropy gives

$$p_j \propto \mu_j e^{-\lambda z_j}$$

(Jaynes 1957; Csiszár 1975).

This construction selects the exponential family only after the score, reference measure, and expected-score constraint have been declared. The multiplier λ is fixed by the supplied numerical constraint m . Without a principle or observation that determines m , maximum entropy does not predict a unique strength.

4.6 The status of λ

The parameter has three distinct statuses that should not be conflated.

Raw-score model

For uncalibrated raw scores, only the product of score scale and coupling strength is identifiable. Numerical λ is representation-dependent.

Standardized-score model

For a fixed nonconstant standardized score vector z , suppose

$$P_\lambda(j) = P_\eta(j)$$

for all j . Taking the ratio of two entries with $z_j \neq z_k$ gives

$$e^{-\lambda(z_j - z_k)} = e^{-\eta(z_j - z_k)},$$

and hence

$$\lambda = \eta.$$

Thus λ is identifiable within the standardized model. Given multinomial transition counts and nonconstant score rows, it can be estimated statistically.

Physical theory

Statistical identifiability is not physical derivation. A fitted parameter can be unique in a declared model while its value remains unexplained by the underlying theory. Neither the variance of a chosen noise law nor the existence of exact coarse-graining fixes λ . Exact exponential aggregation holds for every admissible λ , and a numerical expected-score constraint determines λ only after that constraint is independently supplied.

The safe conclusion is:

Parameter status. The parameter λ may be identifiable and estimable after standardization while remaining underived.

4.7 Context–scale incompatibility

The tension between contextual normalization and cardinal stability can be stated directly. Let $D_S(x, y)$ be the numerical difference between scores assigned to x and y in a finite

context S . Assume:

1. adding arbitrary alternatives leaves $D_S(x, y)$ unchanged;
2. $D(ax + b, ay + b) = D(x, y)$ for every $a > 0$;
3. D is continuous and $D(x, x) = 0$.

Extension stability removes dependence on the surrounding context. Translation invariance gives

$$D(x, y) = h(x - y).$$

Scale invariance gives

$$h(ad) = h(d) \quad \text{for every } a > 0.$$

Hence h is constant on positive separations and constant on negative separations. Continuity at zero and $h(0) = 0$ force both constants to vanish.

Lemma 4.4 (context–scale lemma).

Under the assumptions above, the only continuous cardinal score difference is the trivial one.

The lemma is an elementary consequence of standard interval-scale and meaningfulness arguments (Luce et al. 1990). It does not say that interval-scale differences are meaningless. Ordinary differences transform covariantly with the unit. The incompatibility arises only when one simultaneously demands a nontrivial cardinal difference, complete context-extension stability, and numerical invariance under every positive change of unit.

Raw differences preserve extension stability but need a scale convention. Local standardized scores remove positive-affine scale but depend on context. Ranks are more invariant but ordinal. The framework therefore treats score choice as an explicit modelling contract rather than as a consequence of symmetry alone.

4.8 Kernel-selection boundary

The results of this section establish a hierarchy:

1. q -contrasts can be made operational by a declared kernel;
2. raw exponential kernels have a score-scale/coupling degeneracy;
3. scale-free local principles leave infinitely many kernels;
4. difference-based odds or a maximum-entropy constraint select an exponential family under additional assumptions;
5. neither route determines a unique physical λ .

A coherent operational kernel is therefore possible without circularity, but it is not automatically fundamental. The score representation, admissible choice axioms, reference measure, constraint, and calibration remain part of the model specification.

4.9 Transition to refinement and measure

Every normalized choice or transition model presupposes a rule for how alternatives are weighted when a representation is refined. If one listed alternative is replaced by several exact descriptions, assigning unit coefficient to every listed item changes the normalized law. Kernel selection therefore does not solve multiplicity.

The next section introduces an additive base mass μ to separate transition preference from representational counting. This additional structure is required even when the functional form of the kernel has already been fixed.

5. Measure, Multiplicity, and Refinement

5.1 Why relation plus marks are insufficient

Let a finite marked relational state be represented by

$$(X, R, q),$$

where X is a finite labeled set, R is a relation on X , and $q : X \rightarrow \mathbb{R}$ is a scalar mark. A refinement replaces one macrostate $x \in X$ by several microstates $x_1, \dots, x_m$. If the new states have the same relational profile and the same mark, the refined structure can be interpreted in at least two ways.

In a **descriptive refinement**, the microstates are alternative descriptions of one macrostate. Their total weight should equal the original macro weight. In an **ontic multiplication**, they are distinct states and a counting rule may assign m units of multiplicity. The bare refined relation and the equal q -marks are compatible with both interpretations.

Proposition 5.1 (semantic multiplicity obstruction).

No deterministic rule whose input is only the unweighted marked structure (R, q) can, in general, distinguish a descriptive refinement from an ontic multiplication when the two interpretations yield isomorphic marked structures.

Proof. The input to the rule is identical under the two interpretations, whereas the required macro mass differs. A deterministic function of identical input cannot return both values. $\square$

The proposition does not settle an ontological dispute. It identifies missing information. A refinement map, multiplicity assignment, base measure, or equivalent external semantics must be supplied before the two interpretations can be distinguished.

5.2 Additive base mass

Introduce a positive mass

$$\mu : X \rightarrow (0, \infty).$$

The pair (q, μ) defines a finite marked measure

$$\nu = \sum_{x \in X} \mu_x \delta_{q_x}.$$

Suppose a contextual score $T_\nu(q_x)$ depends only on the marked measure and the marked point. Two simple examples anchor the notation. The choice $T_\nu(q_x) = q_x$ gives a raw-score model. If

$$M = \sum_y \mu_y, \quad \bar{q}_\mu = M^{-1} \sum_y \mu_y q_y, \quad s_\mu^2 = M^{-1} \sum_y \mu_y (q_y - \bar{q}_\mu)^2 > 0,$$

then

$$T_\nu(q_x) = \frac{q_x - \bar{q}_\mu}{s_\mu}$$

gives a standardized contextual score analogous to the row-standardized construction of Section 4.3. Define normalized weights

$$P_x = \frac{\mu_x f(T_\nu(q_x))}{\sum_{y \in X} \mu_y f(T_\nu(q_y))}$$

for a positive function f .

Proposition 5.2 (exact clone-refinement invariance).

Replace a state r by exact clones $r_1, \dots, r_m$ with

$$q_{r_a} = q_r, \quad \sum_{a=1}^m \mu_{r_a} = \mu_r.$$

Then the marked measure ν , every score T_ν , and all macro probabilities are unchanged.

Proof. The clone replacement preserves the atomic measure because

$$\sum_a \mu_{r_a} \delta_{q_r} = \mu_r \delta_{q_r}.$$

The combined unnormalized clone weight is

$$\sum_a \mu_{r_a} f(T_\nu(q_r)) = \mu_r f(T_\nu(q_r)).$$

All other unnormalized weights are unchanged, so normalization gives the same macro probabilities. $\square$

The role of μ is therefore operationally precise: it prevents the representation from treating the number of listed descriptions as automatic physical multiplicity. This does not determine the origin or interpretation of μ .

5.3 Finite refinement trees

Let $\mathcal{T}$ be a finite rooted refinement tree. Assign each terminal leaf ℓ a positive weight w_ℓ .

Lemma 5.3 (unique additive extension on a finite tree).

Every finitely additive extension agreeing with the leaf weights must satisfy

$$\mu(A) = \sum_{\ell \in \text{Desc}(A)} w_\ell$$

for every node A . This extension is unique and independent of the order in which refinements are introduced or merged.

Proof. Repeated finite additivity expresses the mass of a node as the sum of the masses of its children and, recursively, as the sum of all descendant leaf weights. Any two additive extensions agreeing on the leaves therefore agree on every node. $\square$

The same argument applies to additive sufficient statistics. For example,

$$\sum_{\ell} w_{\ell} q_{\ell}, \quad \sum_{\ell} w_{\ell} q_{\ell}^2, \quad \sum_{\ell} w_{\ell} e^{-\lambda q_{\ell}}$$

can be aggregated along any hierarchy without path dependence. Derived quantities such as weighted means or exponential effective scores are path-independent when the additive messages needed to reconstruct them are carried.

Uniform leaf weights recover descendant counting measure, but only conditionally on a specified terminal-leaf interpretation. Additivity does not decide how many terminal states exist or whether a new branch represents a new physical alternative.

5.4 Infinite refinement without terminal leaves

Terminal leaves are not mathematically necessary. Consider a finitely branching rooted tree in which every node A assigns positive fractions

$$p(B \mid A) > 0$$

to its children B , with

$$\sum_{B \prec A} p(B \mid A) = 1.$$

The mass of a cylinder determined by a finite path is the product of its branch fractions. These cylinder probabilities are consistent: the mass of a parent cylinder equals the sum of the masses of its child cylinders. Because a finitely branching rooted tree has a countable cylinder algebra and its path space carries the standard cylinder sigma-algebra, the consistent premeasure extends to a probability measure on the infinite path space (Kolmogorov 1933).

This construction separates two concepts that are easily conflated: terminality and atomicity.

Example 5.4 (nonterminal atom).

Along one infinite path, let the retained branch fraction at depth $n \geq 2$ be

$$p_n = 1 - \frac{1}{n^2}.$$

Then the depth- N cylinder mass is

$$\prod_{n=2}^N \left(1 - \frac{1}{n^2}\right) = \frac{N+1}{2N} \longrightarrow \frac{1}{2}.$$

The infinite path is nonterminal but is an atom of mass $1/2$.

Example 5.5 (terminal-free non-atomic measure).

If every child fraction is bounded above by a fixed $r < 1$, then every depth- d cylinder has mass at most r^d , which tends to zero. No individual infinite path is an atom.

Thus an atom need not be terminal, and the absence of terminal leaves does not imply non-atomicity.

5.5 Projective branch fractions

Let a refinement hierarchy carry a positive finitely additive weight W . For a node A partitioned into children $B_1, \dots, B_k$, define

$$p(B_a \mid A) = \frac{W(B_a)}{W(A)}.$$

These fractions are normalized and telescope along paths. If ℓ is a leaf or a cylinder endpoint,

$$\prod_{A \rightarrow B \text{ on path to } \ell} p(B \mid A) = \frac{W(\ell)}{W(\text{root})}.$$

The result is independent of intermediate grouping. Conversely, a path-independent branching law compatible with disjoint regrouping defines an additive weight up to one common multiplicative constant.

This ratio architecture is ordinary conditional-measure structure. Its significance here is restrictive rather than generative: projective consistency determines how supplied weights must combine, but it does not select the weights.

A principal example used later is

$$W_\lambda(A) = \sum_{\ell \in A} \mu_\ell e^{-\lambda q_\ell}.$$

It is additive, so the ratios $W_\lambda(B)/W_\lambda(A)$ are projective. Section 6 develops the associated effective score and proves its exact static aggregation. The present section uses the example only to show that projectivity acts on additive weights, not on an arbitrarily chosen summary such as the exponential of a subtree mean.

5.6 What measure consistency does not determine

The preceding constructions establish several positive results:

1. additive mass removes naive clone-count dependence;
2. finite refinement is path-independent once terminal weights are supplied;
3. consistent infinite refinement supports a probability measure without terminal leaves;
4. branch probabilities have an exact ratio form once an additive weight is supplied.

They do not answer the prior selection problem. The same tree supports many consistent mass assignments, including uniform, biased, atomic, and non-atomic measures. Likewise, many terminal-weight assignments obey the same projectivity and regrouping laws while generating different branch probabilities.

Accordingly, the report treats μ as a separate informational component. It is not reconstructed from R merely by choosing a convenient order statistic, and it is not derived from q merely by exponentiating the marks. A function $\mu = F(R)$ may be a useful order-based summary, but it cannot break an identifiability limit already present within each order-equivalence class.

The conclusion is:

Conclusion. Refinement consistency constrains the algebra of mass, not the origin of mass.

A physical interpretation of μ , if one exists, requires an operational mapping beyond the formal results established here.

6. Static Exponential Aggregation

6.1 Additive exponential weight

Let a finite marked system carry positive masses μ_i and marks q_i . For a fixed nonzero parameter λ , define the additive exponential weight of a block B by

$$W_\lambda(B) = \sum_{i \in B} \mu_i e^{-\lambda q_i}.$$

For disjoint blocks $B_1, \dots, B_r$,

$$W_\lambda \left(\bigsqcup_{a=1}^r B_a \right) = \sum_{a=1}^r W_\lambda(B_a).$$

Consequently, the ratio

$$p(B_a \mid A) = \frac{W_\lambda(B_a)}{W_\lambda(A)}$$

is normalized for every partition of A , and nested branch probabilities are independent of regrouping. This is the ordinary ratio architecture of an additive positive measure applied to the exponential weight.

The construction is conditional. It assumes that μ , q , and λ have already been supplied. Projectivity constrains how their weights combine; it does not derive them.

6.2 Effective score and block message

Let

$$\mu_B = \sum_{i \in B} \mu_i.$$

When $\mu_B > 0$, define

$$Q_\lambda(B) = -\frac{1}{\lambda} \log \left(\frac{W_\lambda(B)}{\mu_B} \right).$$

Then

$$W_\lambda(B) = \mu_B e^{-\lambda Q_\lambda(B)}.$$

The pair

$$(\mu_B, Q_\lambda(B))$$

is therefore an exact block message for the exponential observable. It contains precisely the information needed to reconstruct the block's total exponential weight.

When the total mass is normalized to one, the global score is

$$Q_\lambda(q, \mu) = -\frac{1}{\lambda} \log \sum_i \mu_i e^{-\lambda q_i}.$$

Up to sign conventions, this is the exponential certainty equivalent or entropic transform (Föllmer and Schied 2002). The identification is mathematical. No thermodynamic energy, temperature, or physical free energy is inferred.

6.3 Exact hierarchical associativity

Partition the microscopic indices into blocks G . Let

$$\mu_G = \sum_{i \in G} \mu_i$$

and define the normalized within-block score

$$Q_G = -\frac{1}{\lambda} \log \left(\frac{1}{\mu_G} \sum_{i \in G} \mu_i e^{-\lambda q_i} \right).$$

Then

$$\mu_G e^{-\lambda Q_G} = \sum_{i \in G} \mu_i e^{-\lambda q_i}.$$

Summing over blocks gives

$$\sum_G \mu_G e^{-\lambda Q_G} = \sum_i \mu_i e^{-\lambda q_i},$$

and hence

$$Q_\lambda(q, \mu) = -\frac{1}{\lambda} \log \sum_G \mu_G e^{-\lambda Q_G}.$$

Proposition 6.1 (exact static aggregation).

Passing each block's total mass and exponential effective score reproduces the global score exactly, independently of the order of grouping.

The proposition is the log-sum-exp or exponential quasi-arithmetic associativity property. Its role in the framework is to identify a sufficient static message for one declared observable. It is not a uniqueness theorem for all coarse-graining problems.

6.4 Gauge and affine covariance

A common shift of all marks gives

$$Q_\lambda(q + c, \mu) = Q_\lambda(q, \mu) + c.$$

For $a > 0$,

$$Q_\lambda(aq + c, \mu) = c + aQ_{a\lambda}(q, \mu).$$

Lemma 6.2 (affine covariance).

The exponential effective score is translation covariant, while positive rescaling of q transports the transform parameter from λ to $a\lambda$.

Proof.

$$\begin{aligned} Q_\lambda(aq + c, \mu) &= -\frac{1}{\lambda} \log \sum_i \mu_i e^{-\lambda(aq_i + c)} \\ &= c - \frac{1}{\lambda} \log \sum_i \mu_i e^{-a\lambda q_i} \quad \square \\ &= c + aQ_{a\lambda}(q, \mu). \end{aligned}$$

This is classical log-Laplace algebra. It shows that the numerical scale of q and the parameter labeling the transform curve are coupled. It does not establish a physical renormalization flow.

6.5 Relative-entropy representation

Let μ be a normalized positive reference measure. Maximizing relative entropy

$$-\sum_i p_i \log \frac{p_i}{\mu_i}$$

subject to normalization and a fixed expected mark yields

$$p_i = \frac{\mu_i e^{-\lambda q_i}}{\sum_j \mu_j e^{-\lambda q_j}}$$

for a Lagrange multiplier λ (Jaynes 1957; Csiszár 1975). The normalizing denominator is exactly W_λ when μ is normalized.

This variational representation explains why the base measure cannot be silently discarded. Different μ define different relative-entropy problems and different normalized laws even when the q -values are unchanged.

The representation is conditional on the chosen constraint and reference measure. It does not select a unique physical expected value, a unique λ , or a unique microscopic interpretation.

6.6 Observable-relative sufficiency

The score Q_λ is sufficient for the exponential observable because

$$\mu_B e^{-\lambda Q_B} = \sum_{i \in B} \mu_i e^{-\lambda q_i}.$$

It is not generally sufficient for another observable. For example, a block mean preserves the first weighted moment, while a minimum preserves an extremal mark. Two blocks can have the same Q_λ and different means, variances, quantiles, or transform values at another parameter.

Even within the exponential family, the score is parameter-specific. Exact aggregation holds for every fixed λ , but the block value $Q_\lambda(B)$ changes with λ . Static closure therefore does not select the transform parameter.

The correct claim is:

Static sufficiency. The pair $(\mu_B, Q_\lambda(B))$ is an exact static message for the declared exponential observable at the declared λ .

It is not “the” unique macrostate of the system.

6.7 From static aggregation to the dynamic question

Static associativity answers a regrouping question: can the same present observable be computed after microscopic states are replaced by block messages? For W_λ , the answer is yes.

A dynamic closure question is stronger: do the block messages contain enough information to determine their own future values under a microscopic update? Nothing in Proposition 6.1 guarantees this. The affine covariance identity already warns that an update involving a scale change can require $Q_{a\lambda}$ even when the present macrostate carries only Q_λ .

This distinction motivates Section 7:

Key distinction. Exact static aggregation does not imply autonomous dynamic closure.

7. Dynamic Transport and Nonclosure

7.1 Exponential effective score

Let $q_1, \dots, q_m \in \mathbb{R}$ carry positive masses $\mu_1, \dots, \mu_m$, and set

$$M = \sum_i \mu_i.$$

For $\lambda \neq 0$, define

$$Q_\lambda(q, \mu) = -\frac{1}{\lambda} \log \left(\frac{1}{M} \sum_{i=1}^m \mu_i e^{-\lambda q_i} \right).$$

Only the normalized weights μ_i/M enter the score. Multiplying every mass by one common positive constant leaves Q_λ unchanged.

Up to sign conventions, this is the exponential certainty equivalent or entropic transform familiar from exponential utility and convex risk functionals (Föllmer and Schied 2002). The present use is formal: no identification with physical free energy is made.

The score is translation covariant:

$$Q_\lambda(q + c, \mu) = Q_\lambda(q, \mu) + c.$$

More generally, for $a > 0$,

$$Q_\lambda(aq + c, \mu) = c + aQ_{a\lambda}(q, \mu).$$

This identity is a direct algebraic property of the log-Laplace transform. Its importance here is that changing the scale of q transports the parameter at which the transform must be evaluated.

7.2 Centered affine contraction

Consider the deterministic update

$$q'_i = \bar{q}_\mu + a(q_i - \bar{q}_\mu), \quad 0 < a < 1,$$

where

$$\bar{q}_\mu = \frac{1}{M} \sum_i \mu_i q_i.$$

Equivalently,

$$q'_i = (1 - a)\bar{q}_\mu + a q_i.$$

Proposition 7.1 (transport identity).

Under this update,

$$Q_\lambda(q', \mu) = (1 - a)\bar{q}_\mu + a Q_{a\lambda}(q, \mu).$$

Proof. Apply affine covariance with scale a and shift $c = (1 - a)\bar{q}_\mu$. $\square$

The proposition is exact, but it is not an independent new transform theorem. It is a model-specific dynamical corollary of classical affine covariance. Its substantive consequence is that a fixed observational value Q_λ at the next step depends on the previous score at $a\lambda$, not only at λ .

7.3 An exact scalar nonclosure witness

We now show that the mean and one fixed- λ score do not generally form an autonomous macrostate.

Take support

$$q \in \{0, 1, 2, 3\},$$

parameter

$$\lambda = \log 2,$$

and contraction factor

$$a = \frac{1}{2}.$$

Define two probability vectors

$$p^+ = \left(\frac{9}{40}, \frac{7}{20}, \frac{1}{8}, \frac{3}{10} \right)$$

and

$$p^- = \left(\frac{11}{40}, \frac{3}{20}, \frac{3}{8}, \frac{1}{5} \right).$$

Both are positive and normalized. Their means are equal:

$$\sum_{k=0}^3 k p_k^+ = \sum_{k=0}^3 k p_k^- = \frac{3}{2}.$$

At $\lambda = \log 2$, we have $e^{-\lambda k} = 2^{-k}$, and direct calculation gives

$$\sum_{k=0}^3 p_k^+ 2^{-k} = \sum_{k=0}^3 p_k^- 2^{-k} = \frac{15}{32}.$$

Therefore

$$Q_{\log 2}(p^+) = Q_{\log 2}(p^-).$$

The centered contraction with $a = \frac{1}{2}$ requires the prior transform at

$$a\lambda = \frac{1}{2} \log 2.$$

At this parameter,

$$\sum_{k=0}^3 p_k^+ 2^{-k/2} = \frac{23}{80} + \frac{\sqrt{2}}{4},$$

whereas

$$\sum_{k=0}^3 p_k^- 2^{-k/2} = \frac{37}{80} + \frac{\sqrt{2}}{8}.$$

The difference is

$$\frac{5\sqrt{2} - 7}{40} > 0.$$

Hence

$$Q_{\frac{1}{2} \log 2}(p^+) \neq Q_{\frac{1}{2} \log 2}(p^-).$$

Applying Proposition 7.1 yields

$$Q'_{\log 2}(p^+) \neq Q'_{\log 2}(p^-).$$

Proposition 7.2 (failure of fixed-score closure).

On the unrestricted class of finite marked distributions, the macrostate

$$(\bar{q}_\mu, Q_\lambda)$$

does not generally determine the next Q_λ under the centered affine contraction.

The witness is exact. It does not rely on numerical optimization, random search, or a finite tolerance. It is also deliberately narrow: it proves nonclosure on the unrestricted finite-distribution class, not on every parametric subfamily.

7.4 Why one transform value is insufficient

The score Q_λ is a single evaluation of the log-Laplace transform. One value imposes one nonlinear constraint on the distribution. Even after the mean is fixed, many finite distributions can satisfy that same constraint while differing at another transform parameter. By contrast, a complete Laplace-transform curve, under its standard domain conditions, can determine a distribution; uniqueness results make clear that a sufficiently rich set of transform values contains qualitatively more information than one evaluation (Lin and Dou 2021).

The cumulant expansion illustrates the same hierarchy. When the expansion is valid,

$$Q_\lambda = \kappa_1 - \frac{\lambda}{2}\kappa_2 + \frac{\lambda^2}{6}\kappa_3 - \frac{\lambda^3}{24}\kappa_4 + \cdots.$$

A fixed value at one λ does not separately determine the cumulants. Mean-variance closure is therefore not exact in general. Higher cumulants contribute at finite λ , and a finite truncation is an approximation unless an invariant family independently forces the higher terms to be functions of finitely many parameters. This is the familiar closure problem: a reduced set of observables can evolve through unresolved information (Kuehn 2016).

7.5 Closure hierarchy

The update admits a precise hierarchy of closure statements.

One fixed score

A single Q_λ , even together with $\bar{q}_\mu$, is not generally closed, as Proposition 7.2 shows.

The full transform curve

If the macrostate carries the mean and the entire function

$$\lambda \mapsto Q_\lambda,$$

then the deterministic contraction is closed:

$$Q'_\lambda = (1 - a)\bar{q}_\mu + aQ_{a\lambda}.$$

The update acts by a shift of value and a rescaling of the curve's argument.

Restricted invariant families

A finite-dimensional closure may exist when the microscopic distribution is known to remain in an invariant parametric family. For example, affine transformations preserve Gaussianity, and for a Gaussian distribution

$$Q_\lambda = \bar{q}_\mu - \frac{\lambda}{2}\sigma^2.$$

In that restricted family, mean and variance determine the whole curve. This conditional exception does not restore general closure; it identifies the extra assumption that makes finite closure possible.

Approximate moment closure

A finite cumulant truncation can be useful for small λ or in regimes where higher cumulants are controlled. It must be labeled as an approximation, not as the exact state of the unrestricted dynamics.

7.6 Static sufficiency is not dynamic autonomy

The same score can be an exact static message and an incomplete dynamic state. For a fixed partition sum,

$$W_\lambda = \sum_i \mu_i e^{-\lambda q_i},$$

the pair $(\sum_i \mu_i, Q_\lambda)$ aggregates exactly across a hierarchy. That static property says that the message preserves one declared observable under regrouping. It does not say that the message contains all information needed by a future update.

This distinction is central:

Nonimplication. Exact static coarse-graining does not imply autonomous macrodynamics.

The result is mathematical and model-conditional. It does not identify Q_λ as a thermodynamic state variable, derive the contraction as a law of nature, or establish a closed physical dynamics.

8. Observable-Relative Coarse-Graining

8.1 Coarse-graining as a preservation problem

Let X be a finite microscopic state space and let

$$\mathcal{P} = \{B_1, \dots, B_r\}$$

be a partition into macrostates. A coarse-graining map is not determined by the partition alone. It must specify which microscopic observables or dynamical predictions are to be preserved.

For a scalar mark q with masses μ , different targets select different messages:

- the weighted arithmetic mean preserves the first moment;
- the exponential effective score preserves an exponential partition sum;
- the minimum preserves an extremal value;
- a quantile preserves a rank-based location but generally not an additive message.

There is therefore no reason to expect one scalar summary to be exactly sufficient for every observable.

8.2 Regular decomposable means

A weighted quasi-arithmetic mean has the form

$$M_f(q; \mu) = f^{-1} \left(\frac{\sum_i \mu_i f(q_i)}{\sum_i \mu_i} \right),$$

where f is continuous and strictly monotone. Such means are hierarchically decomposable when each block transmits its total mass and its block mean.

Classical characterization results of Kolmogorov and Nagumo, with later functional-equation treatments by Aczél, show how regularity, decomposability, and invariance assumptions restrict the generator (Kolmogorov 1930; Nagumo 1930; Aczél 1966). In the translation-covariant class relevant here,

$$M_f(q + c; \mu) = M_f(q; \mu) + c,$$

the regular generators reduce, up to equivalent affine transformations, to the affine family and the exponential family. These yield, respectively, the weighted arithmetic mean and log-sum-exp means.

This characterization does not select one universal mean. It leaves at least two distinct families because they preserve different structures.

8.3 Static messages depend on the observable

Section 6 proved that the pair

$$(M_B, Q_\lambda(B))$$

is an exact hierarchical message for the exponential weight $W_\lambda(B)$. By contrast, the weighted arithmetic mean is exact for the first moment. Neither message is universally sufficient.

The distinction can be stated without repeating the aggregation proof. If two blocks have the same first moment but different exponential weight, an arithmetic-mean message preserves the first target and loses the second. If they have the same Q_λ but different means or different Q_η for $\eta \neq \lambda$, the exponential message preserves only the declared transform value.

Appendix D gives an exact example showing that total mass plus weighted median is not a closed hierarchical message. Robustness, cardinality, and decomposability are therefore separate properties.

8.4 Dynamic aggregation and source occupancy

Now let K be a row-stochastic microscopic transition kernel:

$$K_{ij} \geq 0, \quad \sum_j K_{ij} = 1.$$

For source block A and destination block B , the probability of moving from A to B depends on which microscopic state inside A is occupied. If

$$\rho(i \mid A)$$

is the conditional source occupancy, then

$$K_{AB}^{\text{macro}} = \sum_{i \in A} \rho(i \mid A) \sum_{j \in B} K_{ij}.$$

Two different occupancies on the same microscopic kernel and the same partition can produce different macro rows. Therefore K and $\mathcal{P}$ alone do not generally determine a unique Markovian macro kernel.

There is one important exception. If, for every pair of source states $i, i' \in A$,

$$\sum_{j \in B} K_{ij} = \sum_{j \in B} K_{i'j}$$

for every destination block B , then the macro row is independent of ρ . This is the classical strong lumpability condition (Kemeny and Snell 1960). In the absence of such a condition, occupancy is required.

The obstruction is informational, not metaphysical. A partition tells us which states are grouped; it does not tell us how probability mass is distributed among the grouped source states at the moment of transition.

8.5 Exact flow aggregation

Let a global occupancy distribution π be supplied, with

$$\pi_i \geq 0, \quad \sum_i \pi_i = 1.$$

Define the joint one-step flow

$$F_{ij} = \pi_i K_{ij}.$$

For macro blocks A, B , define

$$\pi_A = \sum_{i \in A} \pi_i$$

and

$$F_{AB} = \sum_{i \in A} \sum_{j \in B} F_{ij}.$$

When $\pi_A > 0$, set

$$K_{AB}^{\text{macro}} = \frac{F_{AB}}{\pi_A}.$$

Proposition 8.1 (exact occupancy-weighted aggregation).

The macro kernel above equals direct microscopic aggregation with the conditional occupancy

$$\rho(i \mid A) = \frac{\pi_i}{\pi_A}.$$

Proof.

$$\frac{F_{AB}}{\pi_A} = \frac{1}{\pi_A} \sum_{i \in A} \sum_{j \in B} \pi_i K_{ij} = \sum_{i \in A} \frac{\pi_i}{\pi_A} \sum_{j \in B} K_{ij}. \quad \square$$

Joint flow is additive under further regrouping, so it provides an exact dynamic message.

However, this positive construction requires π . The mathematics does not supply the initial occupancy, its evolution beyond the declared kernel, or a physical interpretation.

8.6 Static and dynamic sufficiency are different

The static exponential message and the dynamic flow message solve different problems.

For static aggregation, the relevant additive object is

$$W_B = \sum_{i \in B} \mu_i e^{-\lambda q_i}.$$

For dynamic aggregation, the relevant additive object is

$$F_{AB} = \sum_{i \in A, j \in B} \pi_i K_{ij}.$$

Neither can be substituted for the other without further assumptions. A measure μ used to encode refinement multiplicity need not equal a time-dependent occupancy π . A kernel K defines conditional transitions but not the source distribution with which those transitions are averaged.

This yields the organizing principle of the section:

Organizing principle. A coarse-graining variable is exact only relative to a declared observable or dynamics.

The principle does not rule out a richer common state representation capable of preserving several observables simultaneously. It rules out calling one scalar “the” coarse-grained value without specifying the preservation contract.

9. Provisional Informational Architecture

9.1 Obstruction-driven construction

The previous sections do not derive a complete ontology. They identify distinct information deficits that appear when a relational model is asked to support particular operations. The resulting architecture is

$$\mathcal{M} = (R, q, \mu, K, \pi).$$

Each component is introduced for a separate reason:

- R : relational or ordinal structure,
- q : relative cardinal marks,
- μ : additive refinement mass,
- K : conditional transition law,
- π : source occupancy for dynamic aggregation.

The word “provisional” is essential. The tuple is minimal only relative to the operations and representation classes considered here. It is not proved to be the unique smallest description among all possible mathematical formalisms.

9.2 The role of R

The relation R records the primitive relational content admitted by the model. Depending on the application, it may encode reachability, comparability, adjacency, precedence, or another binary structure. From R one may derive invariant ordinal observables such as chains, intervals, quotients, and automorphism classes.

Order-only identifiability shows what R does not determine. Strictly monotone recalibration can leave R unchanged while altering latent marginal scales and densities. Therefore a preferred cardinal coordinate, sampling measure, or numerical spacing cannot be introduced merely by relabeling an order statistic as a metric quantity.

The architecture does not claim that every relation must be primitive. In a more complete theory, R might be generated dynamically or derived from another object. The present point is informational: once R is the observed input, its equivalence classes limit what can be recovered from it.

9.3 The role of q

The mark q supplies cardinal contrasts not measurable from R alone. It is introduced as additional information, not as a theorem that every relational system possesses such a scalar field.

Only relative q -structure is used without calibration. If the operational constructions depend on differences or centered values, then the global shift

$$q_i \mapsto q_i + c$$

is redundant. This is a gauge contract arising from the definitions. It does not establish a physical gauge symmetry.

The scale of q is a separate issue. In exponential kernels, a rescaling of q can be absorbed into the coupling parameter before a standardization convention is fixed. After standardization, the remaining dimensionless parameter may become statistically identifiable, but it remains underived. Thus q adds cardinal information while leaving calibration as an open problem.

9.4 The role of μ

The mass μ supplies additive multiplicity and refinement semantics. Without it, exact clones can change normalized weights merely by changing the number of listed alternatives. Relation plus equal marks cannot decide whether a repeated representation is descriptive or ontically multiplicative.

Once μ is supplied, exact conservative refinements preserve the marked measure

$$\nu = \sum_i \mu_i \delta_{q_i},$$

and additive statistics become path-independent. Consistent branch fractions take the ratio form $W(B)/W(A)$, but consistency does not select the underlying terminal weights or split fractions.

The architecture therefore distinguishes the existence of an additive measure from the origin of that measure. A later empirical or physical theory would need to say what

operational procedure assigns μ .

9.5 The role of K

The kernel K specifies conditional transitions. It is not determined by the existence of R , q , and μ alone. Locality and invariance principles can restrict a family of kernels without selecting a unique member. Additional axioms, calibration, data, or a dynamical law are required.

The exponential form is useful because q -contrasts appear directly in log-odds and because the corresponding partition sums admit exact static messages. These properties make it a mathematically coherent candidate. They do not prove that the kernel is fundamental or physical.

The dynamic nonclosure result further separates the microscopic kernel from a putative scalar macrostate. Exact static aggregation of an exponential score does not guarantee that the same score evolves autonomously.

9.6 The role of π

The occupancy π records how probability mass is distributed over source states at the time of aggregation. A microscopic kernel and a partition do not generally determine a unique macro transition, because averaging a row of K requires source weights. Strong lumpability can remove this dependence, but it is a restrictive structural condition rather than a generic fact.

By carrying

$$F_{ij} = \pi_i K_{ij},$$

one obtains an additive flow that aggregates exactly. This explains why π is not redundant with K : K is conditional, whereas π specifies the condition under which transitions are sampled. Nor is π automatically identical to μ . A refinement mass and a dynamical occupancy can coincide in a special model, but the formal roles are different.

9.7 Non-reduction statements

Within the representation classes studied, the following reductions are not available without extra assumptions.

$$R \not\Rightarrow q$$

Order-preserving transformations can alter latent cardinal quantities while leaving R unchanged.

$$(R, q) \not\Rightarrow \mu$$

Marked relational clones admit incompatible multiplicity interpretations.

$$(R, q, \mu) \not\Rightarrow K$$

Symmetry, locality, and standardized scores leave multiple admissible transition laws unless stronger choice principles or empirical calibration are added.

$$(K, \mathcal{P}) \not\Rightarrow K^{\text{macro}}$$

A partitioned microscopic kernel does not determine a unique macro kernel without occupancy or lumpability.

These are conditional non-reduction statements. They do not prove logical independence in every conceivable formalism, and they do not forbid a future theory from deriving several components from a common deeper object.

9.8 Obstruction map

The architecture can be summarized as a dependency map:

monotone-order ambiguity	$\implies$	additional cardinal structure q ,
clone/multiplicity ambiguity	$\implies$	additive mass μ ,
kernel underdetermination	$\implies$	explicit transition law K ,
scalar dynamic nonclosure	$\implies$	richer macrostate or invariant-family assumption,
macro-transition ambiguity	$\implies$	occupancy π or lumpability.

Some arrows summarize classical mathematics; others summarize project-specific exact identities and counterexamples. The contribution of the map is organizational: it keeps these logical roles separate and prevents a successful construction in one layer from being promoted into a conclusion about another.

9.9 Relative minimality

A precise minimality claim must name its comparison class. The tuple (R, q, μ, K, π) is sufficient to formulate the operations emphasized here:

- ordinal relational observables from R ;
- relative cardinal comparisons from q ;
- conservative refinement and projective weighting from μ ;
- microscopic conditional evolution from K ;
- occupancy-weighted dynamic aggregation from π .

Removing a component reintroduces at least one documented obstruction within the present construction. This is a useful relative-minimality statement. It is not a uniqueness theorem for mathematical ontology.

Alternative representations may package the same information differently. For example, one could carry the joint flow F instead of the pair (K, π) when only one-step aggregation is required. A measure-valued state may combine q and μ . A richer stochastic process may derive occupancy from its history. Such reformulations should be compared by information content and operational sufficiency, not by symbol count.

9.10 Boundary and open bridges

The architecture remains formal. The current results do not derive:

- a physical spacetime;
- calibrated temporal or spatial distances;

- curvature or gravitational dynamics;
- the physical origin of q ;
- a unique or empirically measured μ ;
- a fundamental value of λ ;
- a preferred microscopic kernel K ;
- an empirical occupancy law π .

The absence of these bridges is not hidden in terminology such as “effective geometry” or “free energy.” Any future physical interpretation must specify a measurement map from empirical data to the formal components and must survive prospective tests against alternatives and null models.

The current achievement is more limited but more secure:

Current result. The model’s informational requirements are stated explicitly, and the known closures and nonclosures are separated by operation.

This architecture provides a controlled starting point for further theory. It does not complete that theory.

10. Scope, Contribution, and Conclusion

10.1 What has been established

The report began with a relational representation and asked which additional informational structures are required for a specified set of operations. The answer is not a universal metaphysical inventory. It is a conditional dependency map.

Within the representation classes studied:

1. an order R does not identify latent quantities that vary under order-preserving recalibration;
2. a mark q adds information only when its marked law and observation coupling are not determined by R ;
3. relation plus marks do not fix refinement multiplicity, motivating an additive mass μ ;
4. symmetry and locality can restrict but do not uniquely determine a transition kernel K ;
5. the exponential score Q_λ is an exact static message for one declared exponential observable;
6. one fixed- λ score is not generally a closed dynamic state under the centered contraction;
7. a partitioned microscopic kernel does not generally define a macro kernel without occupancy π or lumpability.

These statements motivate the provisional architecture

$$\mathcal{M} = (R, q, \mu, K, \pi).$$

Its relative minimality is operational: removing a component reintroduces at least one documented obstruction for the tasks considered. Equivalent representations may package the same information differently. For one-step dynamics, for example, joint flow can replace the pair (K, π) .

10.2 Contribution hierarchy

The mathematical ingredients have unequal novelty and should be described accordingly.

Classical or attributed results include monotone and copula invariance, interval-scale arguments, Luce/logit representations, exponential certainty-equivalent algebra, additive measure extension, quasi-arithmetic mean characterizations, moment-closure background, transform uniqueness, and Markov lumpability.

Direct framework corollaries include the operational interpretation of q -contrasts for a declared kernel, the raw-score/coupling degeneracy, exact exponential projectivity, and the centered-contraction transport identity.

The strongest self-contained project result is the exact nonclosure counterexample: equal mean and equal Q_λ do not determine the next Q_λ on the unrestricted finite-distribution class. The broader framework-level contribution is the obstruction map that separates the roles of relation, mark, mass, transition, and occupancy.

10.3 What has not been established

No result in this report derives:

- a physical spacetime or causal structure from the formal relation;
- a calibrated temporal or spatial metric;
- curvature, gravity, or matter dynamics;
- the physical origin or measurement procedure for q ;
- a uniquely preferred mass μ ;
- a fundamental kernel K ;
- a predicted numerical value of β or λ ;
- a physical occupancy law π ;
- a universal scalar coarse-graining variable.

The terms “gauge,” “effective score,” and “flow” are used mathematically. They do not by themselves establish a corresponding physical field theory.

10.4 Open bridges

A physical or empirical extension would require at least four additional bridges.

First, an observation map must specify how data determine or constrain R, q, μ, K, π . Second, calibration standards must separate representation-dependent scale from measurable quantities. Third, proposed dynamics must be tested prospectively against alternative kernels, distribution families, and null models. Fourth, any geometry constructed from the formal variables must be validated independently before receiving a spacetime interpretation.

Mathematically, several questions remain open. The minimal sufficient dynamic state for broader update classes may be smaller than a full transform curve but larger than a single score. Restricted invariant distribution families may admit exact finite closure. Approximate closures require error bounds and domain conditions. Multi-observable coarse-graining may be formulated through vector or measure-valued messages rather than a single scalar.

10.5 Conclusion

The principal lesson is methodological. Relational simplicity at the microscopic level does not justify silently importing cardinal scale, multiplicity, transition law, or occupancy at

later stages. Each layer should be either derived under explicit assumptions or declared as additional information.

The architecture

$$(R, q, \mu, K, \pi)$$

makes those declarations visible. It identifies where exact constructions are available, where classical theorems apply, and where closure fails. That is a more limited conclusion than a completed physical theory, but it is also a testable and auditable foundation on which further work can be built.

Appendix A. Equivariance and External Calibration

A.1 Endogenous equivariance

Let a group G act on a state space $\mathcal{X}$ and observation space $\mathcal{Y}$. Let $P(x, \cdot)$ be a Markov transition kernel on $\mathcal{X}$, and let $O : \mathcal{X} \rightarrow \mathcal{Y}$ be an observation map. Assume

$$P(gx, gA) = P(x, A)$$

for every measurable $A \subseteq \mathcal{X}$ and every $g \in G$, and

$$O(gx) = gO(x).$$

If two initial states differ only by the group action, their evolved laws remain related by that action at every finite time.

Proposition A.1 (equivariant evolution preserves equivalence).

Purely endogenous evolution whose initial law, transition kernel, and observation map are equivariant cannot select a preferred representative from one group orbit.

Proof. The one-step statement follows directly from kernel equivariance. Induction gives the same relation for every finite-time law. Applying the equivariant observation map preserves the orbit relation in observation space. $\square$

The result is conditional. Non-equivariant boundary conditions, calibrated inputs, asymmetric noise laws, or new observables can break the equivalence.

A.2 A calibrated witness

Let x and gx be observationally equivalent under the original relational observation. Introduce an additional mark $c : \mathcal{X} \rightarrow \mathbb{R}$ satisfying

$$c(gx) \neq c(x)$$

for at least one pair in the orbit, and suppose c is externally calibrated rather than computed equivariantly from the original relation. Then the enriched observation

$$\tilde{O}(x) = (O(x), c(x))$$

distinguishes that pair.

Proposition A.2 (calibration can break degeneracy).

An additional non-invariant calibrated observable can separate states that are indistinguishable under an equivariant order-only observation.

This is a constructive possibility, not a derivation of the required calibration. The main text therefore treats calibrated marks as possible bridges rather than as consequences of relational dynamics.

Appendix B. Nondegeneracy Boundary of Weighted Standardization

Let positive masses μ_i have total mass

$$M = \sum_i \mu_i.$$

Define

$$\bar{q}_\mu = \frac{1}{M} \sum_i \mu_i q_i$$

and

$$s_\mu^2 = \frac{1}{M} \sum_i \mu_i (q_i - \bar{q}_\mu)^2.$$

When $s_\mu > 0$, the standardized values are

$$z_i = \frac{q_i - \bar{q}_\mu}{s_\mu}.$$

B.1 Continuity inside the nondegenerate domain

For a fixed context with $s_\mu > 0$, adding a bounded-score alternative of mass ε changes M , $\bar{q}_\mu$, s_μ , the standardized scores, and any continuous normalized positive kernel continuously as $\varepsilon \rightarrow 0$.

Thus conservative small-mass refinements are stable away from the singular boundary.

B.2 Failure of uniform stability

The convergence is not uniform as the original variance tends to zero. At

$$s_\mu = 0,$$

all marks coincide and the standardized representation is undefined. Moreover, an arbitrarily small mass at a separated score can generate order-one standardized contrasts.

For example, begin with unit mass at $q = 0$, and add mass $\varepsilon > 0$ at $q = 1$. Then

$$\bar{q}_\mu = \frac{\varepsilon}{1 + \varepsilon}, \quad s_\mu^2 = \frac{\varepsilon}{(1 + \varepsilon)^2}.$$

The standardized values are

$$z_0 = -\sqrt{\varepsilon}, \quad z_1 = \frac{1}{\sqrt{\varepsilon}}.$$

Although the added mass tends to zero, its standardized score diverges. A nonlinear kernel of z can therefore change by order one or more near the zero-variance boundary.

Proposition B.1 (standardization boundary).

Weighted standardization is continuous under vanishing bounded mass on every domain bounded away from zero variance, but it is undefined at zero variance and is not uniformly stable as variance approaches zero.

The consequence is local: standardized contextual kernels require an explicit nondegeneracy domain, a regularization rule, or a separate boundary definition. It does not imply that all weighted coarse-graining is unstable.

Appendix C. Restricted Closure and Transform Domain

C.1 Gaussian invariant-family exception

Consider

$$X_{t+1} = b + aX_t + \xi_t,$$

where X_t is Gaussian and ξ_t is independent Gaussian noise. Then X_{t+1} is Gaussian with

$$m_{t+1} = b + am_t, \quad v_{t+1} = a^2v_t + v_\xi.$$

For a Gaussian law,

$$Q_\lambda = -\lambda^{-1} \log E[e^{-\lambda X}] = m - \frac{\lambda}{2}v.$$

Therefore mean and variance determine the full transform curve inside the Gaussian invariant family.

Proposition C.1 (conditional Gaussian closure).

Mean-variance closure is exact under affine dynamics when both the state and independent innovations remain Gaussian.

This exception does not contradict the general nonclosure counterexample. It adds an invariant-family assumption that removes the unresolved higher cumulants. Non-Gaussian innovations regenerate higher cumulants according to

$$\kappa_r(X_{t+1}) = a^r \kappa_r(X_t) + \kappa_r(\xi_t), \quad r \geq 2.$$

Centering is a linear operation and does not by itself make a non-Gaussian innovation law Gaussian.

C.2 Exponential-moment domain

The existence of a finite variance does not guarantee that

$$E[e^{-\lambda X}]$$

is finite for nonzero λ . Some heavy-tailed distributions have finite variance but no moment-generating function away from the origin.

Proposition C.2 (domain warning).

The score Q_λ is defined only on the side of the origin where the corresponding exponential moment is finite. Finite variance alone is insufficient.

The report therefore uses finite-support distributions for the exact nonclosure witness and states transform-domain assumptions whenever an infinite law is discussed. Weak convergence or a central limit theorem also does not, by itself, imply convergence of exponential moments; uniform integrability or comparable tail control is required.

Appendix D. A Nondecomposable Median Message

Robustness of a statistic does not imply hierarchical decomposability. Consider weighted medians under the convention that the median is the smallest value whose cumulative mass reaches at least one half of total mass.

Define two blocks of total mass one:

$$A : \quad 0 \text{ with mass } \frac{3}{5}, \quad 100 \text{ with mass } \frac{2}{5},$$

and

$$A' : \quad 0 \text{ with mass } \frac{3}{5}, \quad 1 \text{ with mass } \frac{2}{5}.$$

Both blocks have total mass 1 and weighted median 0. Let a third block be

$$B : \quad 50 \text{ with mass } \frac{1}{2}.$$

For $A \cup B$, the ordered masses are

$$0 : \frac{3}{5}, \quad 50 : \frac{1}{2}, \quad 100 : \frac{2}{5}.$$

The total mass is $3/2$, so the half-mass threshold is $3/4$. The cumulative mass at 0 is $3/5 < 3/4$, while at 50 it exceeds the threshold. Hence the combined median is 50.

For $A' \cup B$, the ordered masses are

$$0 : \frac{3}{5}, \quad 1 : \frac{2}{5}, \quad 50 : \frac{1}{2}.$$

The cumulative mass first reaches $3/4$ at 1. Hence the combined median is 1.

Proposition D.1 (mass plus median is not closed).

Two blocks can have the same total mass and weighted median yet yield different global medians when combined with the same third block. Therefore the pair “mass plus weighted median” is not an exact hierarchical message.

The result does not rule out mergeable quantile sketches, interval-valued summaries, or richer order-statistic messages. It rules out exact closure of this particular scalar summary.

Report Series, Data, and Code Availability

This document is issued as **Modal Field Research Programme Technical Research Report MFRP-TR-2026-01**. Later reports may reuse the programme-level architecture, but each report must state its own assumptions, claims, and validation status.

Supporting protocols, exact witnesses, claim maps, and validation scripts are available in the public repository:

<https://github.com/DevHabito/precategorical-modal-field-framework>

The exact four-point nonclosure witness is included in the repository’s `audits` directory. The principal arguments in the report are analytic; the computational material supplies reproducibility and regression checks.

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